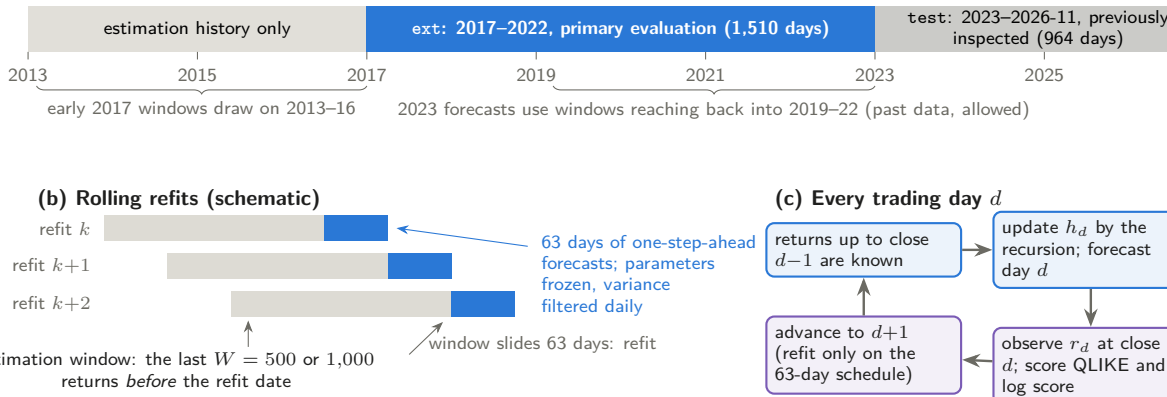

(a) Data and evaluation segments (daily returns from 2013-01-03)
estimation history only
ext: 2017–2022, primary evaluation (1,510 days)
test: 2023–2026-11, previously inspected (964 days)
2013
2015
2017
2019
2021
2023
2025
early 2017 windows draw on 2013–16
2023 forecasts use windows reaching back into 2019–22 (past data, allowed)
(b) Rolling refits (schematic)
refit k
refit $k+1$
refit $k+2$
63 days of one-step-ahead forecasts; parameters frozen, variance filtered daily
estimation window: the last $W = 500$ or $1{,}000$ returns *before* the refit date
window slides 63 days: refit
(c) Every trading day d
returns up to close $d-1$ are known
update h_d by the recursion; forecast day d
observe r_d at close d; score QLIKE and log score
advance to $d+1$ (refit only on the 63-day schedule)